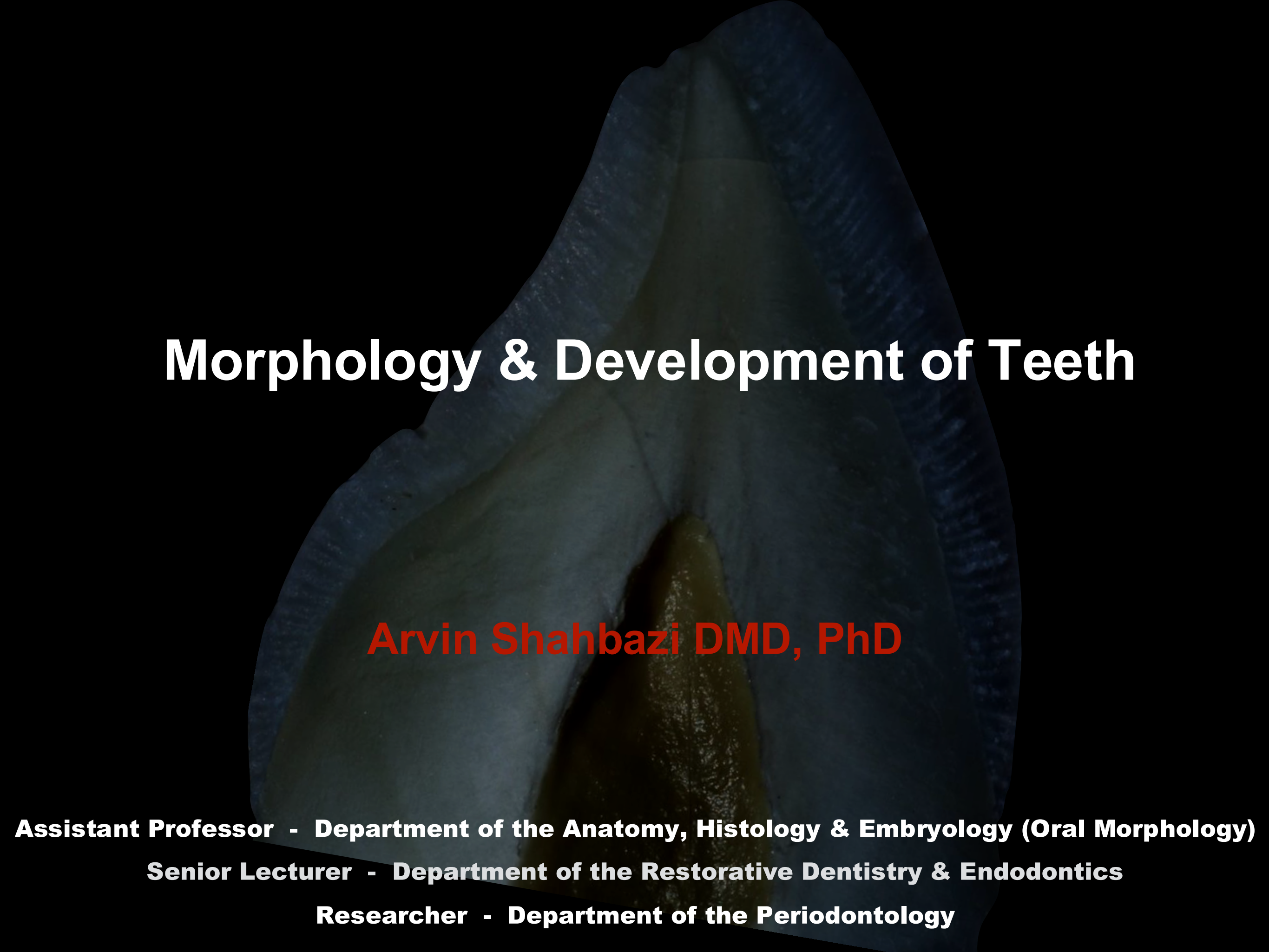




Morphology & Development of Teeth

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Morphology of teeth

Enamel:

Hardest tissue

Covers outer part of the [tooth](#)

Mainly made by hydroxyapatite, calcium & phosphate

Dentin:

A hard tissue containing dental tubules

Located under the enamel

If it is damaged → heat or cold can pass through dental tubules & cause sensitivity or pain

Pulp:

Soft tissue (life of the tooth)

Vessels & nerves are present

Cementum:

Connective tissue

Covers the roots of the teeth

Connects firmly to the gingiva & the alveolar bone

Periodontal ligament:

Connective tissue holding the roots tightly in the alveolar socket



Morphology of teeth

Crown

Anatomical crown

The part of the tooth covered
by enamel

Clinical crown

The part of the tooth visible in
the oral cavity

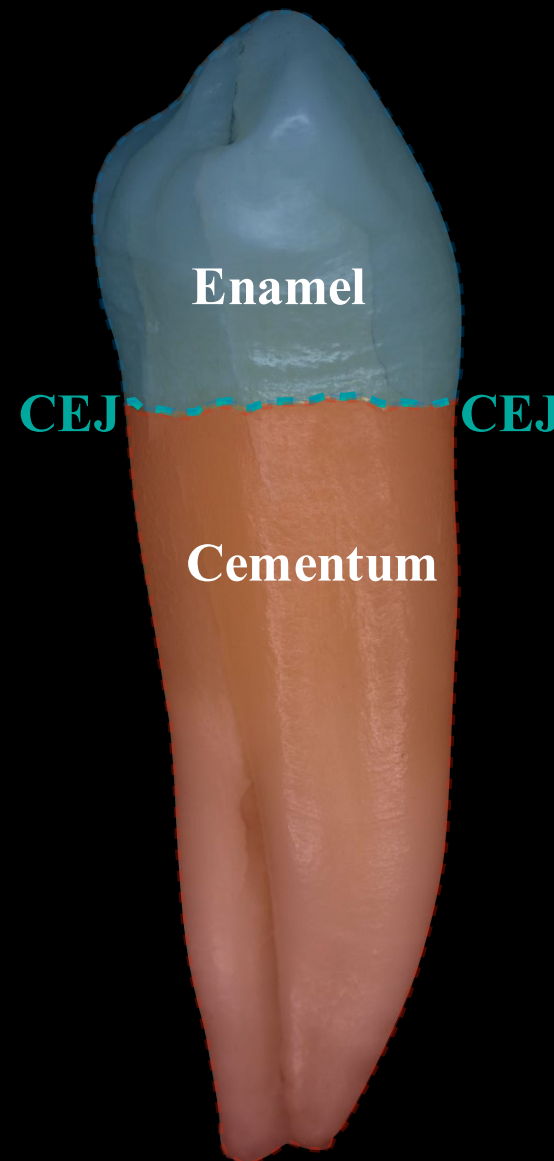
Root

Anatomical root

The part of the tooth covered
by cementum

Clinical root

The part of the tooth lies
within the alveolus



Cementoenamel junction (CEJ)

(Neck of the tooth)

The part of the tooth where 2 vital protecting
substances (enamel + cementum) meet

Numbers & Orientations of teeth

Number of teeth in adults: 32

In each quadrant 8 teeth are present:

1 Central Incisor

1 Lateral Incisor

1 Canine

2 Premolars (bicuspid)

3 Molars (last molar named as Wisdom)

Number of teeth in children: 20

In each quadrant 5 teeth are present:

1 Central Incisor

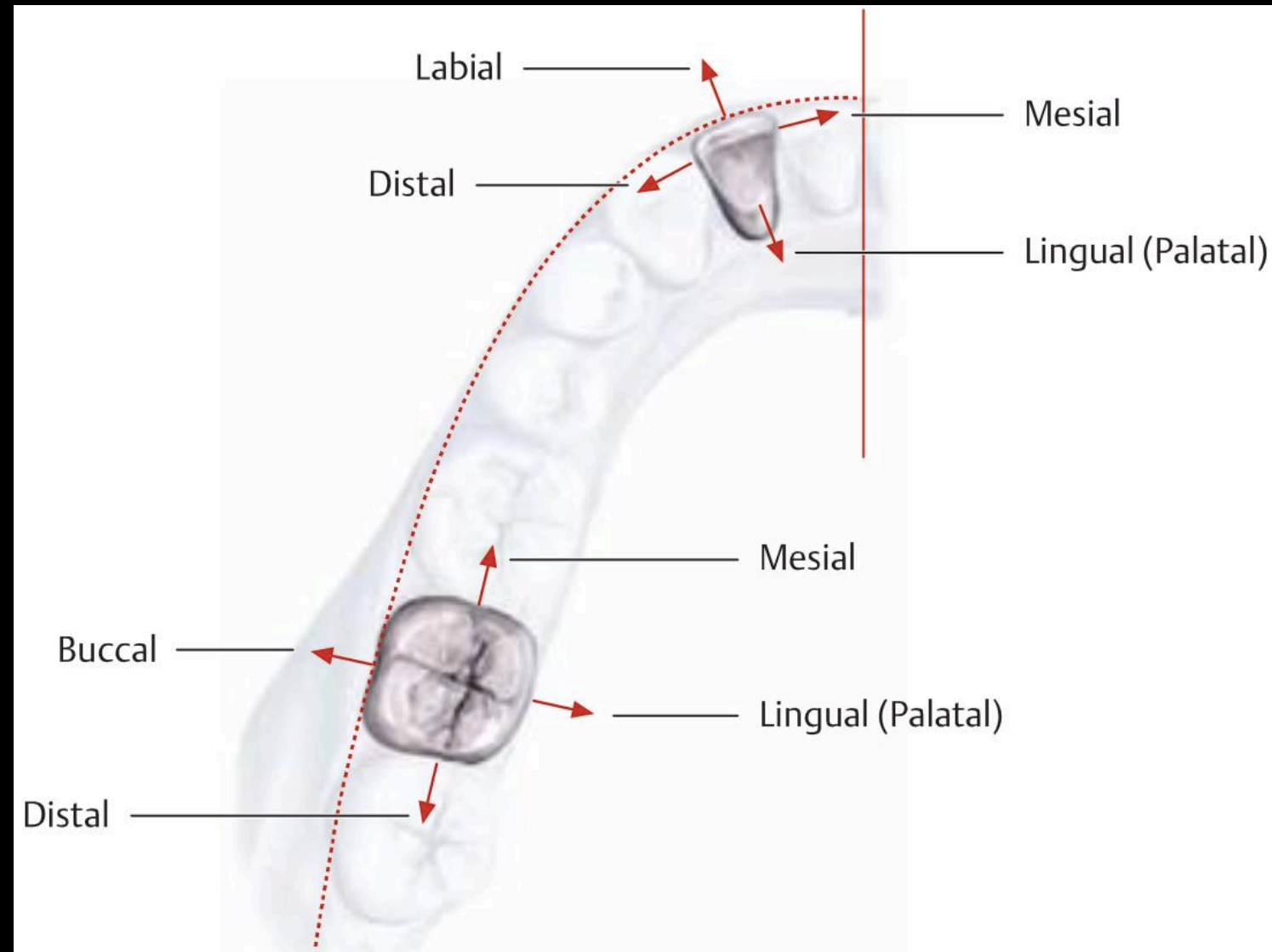
1 Lateral Incisor

1 Canine

2 Molars

No premolars

No wisdom teeth



Maxillary first (central) permanent incisor



- **Crown:**

The shape is similar to the shovel or chisel

- **Cervical section:**

Circular (also triangular) shape

- **Root:**

1 root – 1 rootcanal



Maxillary second (lateral) permanent incisor



- **Crown:**

Smaller, but similar to the 1st incisor

Note: The *mesial angle is rounded*

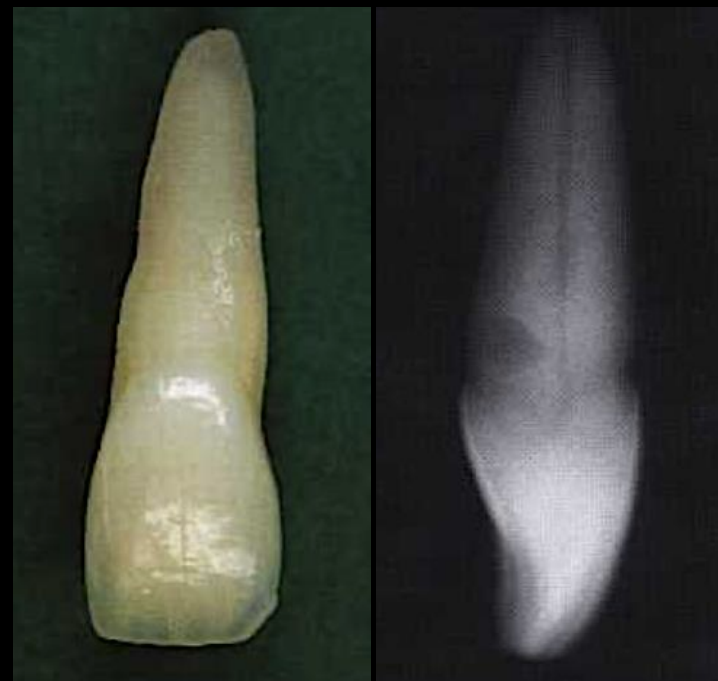
Foramen coecum is on the **palatal** surface

- **Cervical section:**

Flattened in mesiodistal direction (or can be circular shaped)

- **Root:**

1 root – 1 root canal



**In many cases this tooth
might be missed (aplasia)!**



Maxillary permanent canine

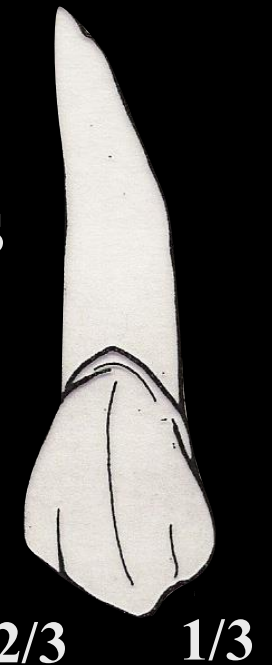
- **Crown:**

Wedge shaped

The edges beginning from tip of the cusp divides the vestibular coronal surface into two parts:

1/3 — smaller part

2/3 — bigger part



- **Cervical section:**

Rounded equilateral

- **Root:**

1 root — 1 rootcanal

It has the longest root (20-22 mm) among the teeth



Mandibular first (central) permanent incisor



- **Crown:**

Chisel shaped

Smallest tooth of the oral cavity!

- **Cervical section:**

Elliptic

- **Root:**

1 root – 1 rootcanal (1/4 contain 2 root canals)



Mandibular second (lateral) permanent incisor



- **Crown:**

Chisel shaped & bit bigger than the 1st lower incisor

- **Cervical section:**

Rectangular with rounded angles

- **Root:**

1 root – 1 rootcanal



Mandibular permanent canine



- **Crown:**

Similar to the upper canine, but smaller & rounded shaped

- **Cervical section:**

Ellipsoid shaped

- **Root:**

1 root – 1 root canal

It has the second longest root among the teeth!



In **1.5 %** of the cases the apex of the root
might be bifurcated!



Maxillary first premolar (bicuspid)

- **Crown:**

2 cusps: 1 buccal - 1 palatal

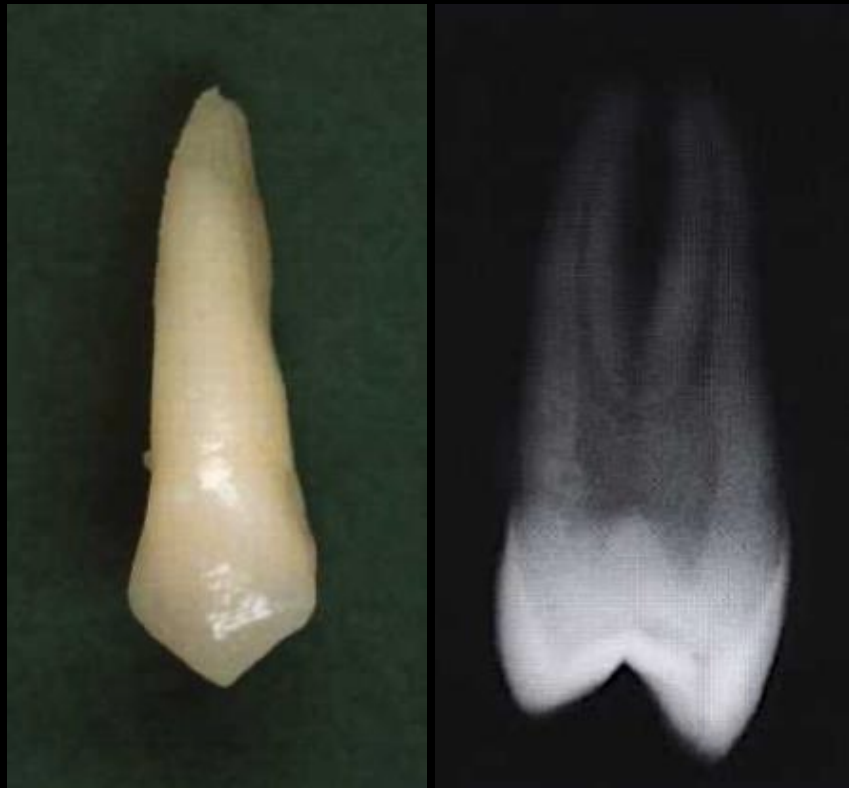
Mesial surface is concave!

- **Cervical section:**

Finger biscuit

- **Root:**

2 roots – 2 canals: 1 buccal, 1 palatal





Maxillary second premolar (bicuspid)

- **Crown:**

It is smaller & similar to the 1st premolar

The cusps are totally the same

- **Cervical section:**

Irregularly flattened

- **Root:**

1 root – 1 rootcanal

Rarely it has 2 canals or 2 roots!

Mandibular first premolar (bicuspid)



- **Crown:**

2 cusps – 1 buccal, 1 lingual

Buccal cusp bigger than the lingual cusp.

The occlusal surface **diverges** to the lingual surface.

- **Cervical section:**

Irregular flattened shape (in mesiodistal direction)

- **Root:**

1 root – 1 canal



Mandibular second premolar (bicuspid)

- Crown:

It is bigger than the lower first premolar

It may have 2 / 3 cusps:

1 buccal cusp - 1 or 2 lingual cusps

- Root:

1 root – 1 canal



Maxillary first molar

■ Crown:

4 cusps

MB (mesiobuccal)

DB (distobuccal)

MP (mesiopalatal)

DP (distopalatal)

MP cusp is the **biggest**, and it has a special cusp:
tuberculum anomale Carabelli

Between MP-DB cusps, there is a projection named as
crista transversa

DP is the **smallest**

There is foramen coecum on the palatal surface of the tooth

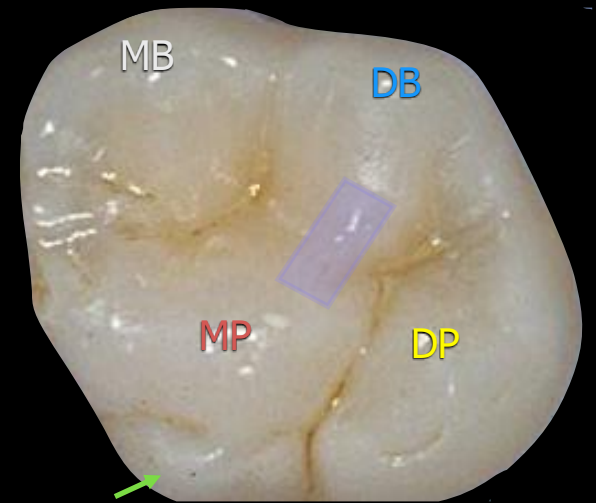
■ Root:

3 roots – 3 or 4 canals

MB – 1 or 2 canals

DB - 1 canal - it is the weakest root!

P - 1 canal – it is the biggest root!



B-view

M-view

P-view

D-view

Maxillary second molar



■ Crown:

Smaller than the 1st molar

Shape is variable

■ Root:

3 root – 3 rootcanal

MB

DB

P

Maxillary third molar

■ Crown:

The shape is variable!

2-6 cusps!

■ Root:

Number & form of roots
are variable too!



Mandibular first molar

■ Crown:

5 cusps:

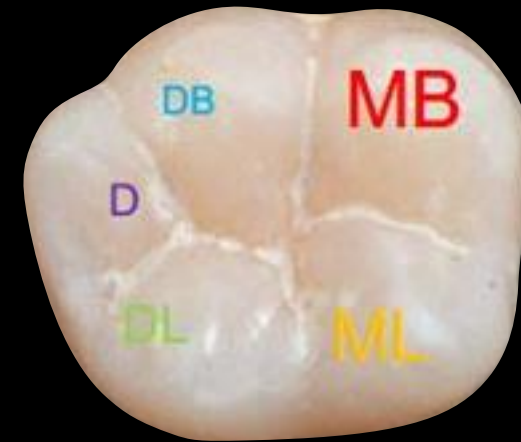
MB (mesiobuccal)

DB (distobuccal)

D (distal)

ML (mesiolingual)

DL (distolingual)



MB cusp is the largest!

DB cusp is the smallest!

Among the cups there are fissures & fossa centralis

There is foramen coecum on the buccal surface

■ Root:

2 roots - Mesial & distal roots

M root – 2 canals: MB, ML canals

D root – 1 canal



Mandibular second molar

- **Crown:**

4 cusps: MB, DB, ML, DL

There are 2 fissures - perpendicular to each other

There is foramen coecum on the buccal surface

- **Root:**

2 roots: Mesial & Distal roots

M root – 1 or 2 canal(s)

D root – 1 canal



Mandibular third molar

- **Crown:**

Smaller than the lower 2nd molar

Sometimes it has 4 or 5-6 cusps

- **Root:**

Usually it has 2 roots:

Mesial & Distal roots, or

sometimes it can have 1-4 roots!

Not just tooth morphology, but also tooth positioning can be crucial in **criminal situations (forensic odontology)**

